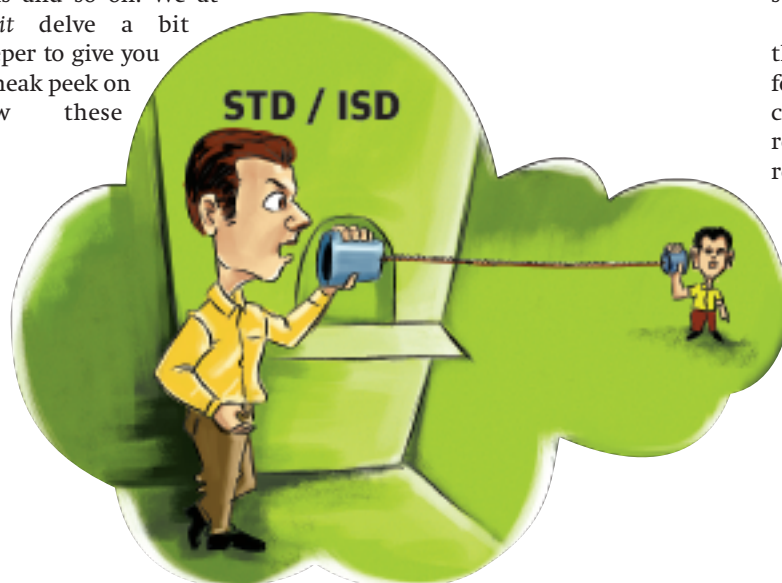


EMERALD

Dreams

Electricity is one of the most important sources of energy that drives us. According to International Energy Outlook 2008, an official energy statistical release by the US government, global electricity generation will nearly double from about 17.3 trillion kilowatt-hours in 2005 to 24.4 trillion kilowatt-hours in 2015 and 33.3 trillion kilowatt-hours in 2030. Even though nuclear power and other means of electricity generation will increase over the period, the most carbon-intensive source—coal—will still dominate, says the report.

Last month, we carried a news snippet underscoring the research initiatives for harnessing mechanical energy into electrical energy that can be used to power your portables. However, on a larger scale there are many initiatives being undertaken to reduce the carbon footprints in the domains of computing, and daily lifestyles in general. It is not only computing that matters, there are these all new breed of hybrid vehicles and electric cars, organic fuels including bio-diesels and natural gases, fuel cells and so on. We at Digit delve a bit deeper to give you a sneak peek on how these



Dragging your processors 24x7? Overclocking your machines? Paying soaring power bills? Read through this section to find innovative researches and their implementation that claims to reduce your costs to nearly one fourth!

greener initiatives will change the face of our technology and reduce our carbon footprints.

As the transistor count of today's microprocessors increases, overheating and power dissipation is becoming a vital issue. As we move towards growing transistor microprocessors, the increasing power budgets of these chips must be addressed. Research carried out at Purdue suggests that after exceeding 35 to 40 W, additional power dissipation increases the total cost per CPU chip by more than a dollar per watt.

Efficient Cooling Systems

Research initiatives for efficient cooling systems are inevitable. Especially when current technologies available for cooling cannot break beyond 200 watts of heat per square centimetre. A Purdue university research team led by Professor Issam Mudawar not only managed to break this barrier, but exceeded it by over five times when they created a way to cool chips that generate more than 1,000 watts of heat per square centimetre.

A fan is an almost inevitable component of the same. However, even if these are sufficient for basic computer applications like business computing and Net surfing, this method is rendered ineffective when the machines are revved by gamers and overclockers.

The new technology is based on a water cooling system as against the conventional air cooled ones. The method implements series of microjets that distributes a coolant in miniscule channels over the top of the chip. The coolant in this case is a hydrofluorocarbon. The heat generated by the chips cause these narrow grooves to heat up, which in turn causes the coolant liquid to vapourise.

One of the biggest advantages of liquid cooling over the conventional air cooled methods is the uniform pattern of cooling

seen in the former. The reason being, while the airflow cannot be kept under control, coolant liquid jets can be maintained uniformly along the length of micro channels. Uniform cooling prevents formation of any hotspots and increases the life of the processors.

Mudawar explains that it is not the superior properties of the coolant that results in such a significant improvement in cooling mechanism, but it is the hybrid nature of the overall mechanism. It's hybrid because it relies on two cooling methods: greater surface area via the microchannels and better circulation via the microjets. The coolant collects at both ends of the channels and is circulated back through the system.

Mini Computers And Applications

Cloud computing is the new buzzword. Not sure what the term connotes? Just turn over the pages to find it explained in the W5H section this issue. Now the question arises—what makes it so popular? Nothing else but greater economy and flexibility.

Zonbu is just one of the products being churned out that function on this concept. In the wake of technological initiatives, products like Zonbu are power miser alternatives that business computing demands. Zonbu works on a Linux platform and considering the minimalist apps that it carries, it doesn't even resort to a fan for its cooling needs. A mini device, no moving parts, energy efficient components and that's all that an Outlook and MS Office user would ever require. On board flash memory as opposed to hard disk based one further reduces its power requirements.

All it requires is a meagre 10-watt power supply and it performs as well as any laptop can perform for the minimal usage that it allows. It can play music, movies, check email, and comes with the Open Office suite. A gamer of course won't go for these mini apps, but a business user and Net surfer wouldn't demand for anything more.

For a comparable output, most conventional CPUs will consume 60-100 watts or more—approximately 4 to 5 times more than what Zonbu does. Zonbu authorities claim that it'll save more than 1,200 kilowatt hours over the course of a year. If you prefer to be more conservative, you can also add on an optional solar panel to power the device for mini applications.



Task Specific Computing

As computing gets more and more task specific, the micro architecture and microprocessors that go into making a computer also are drastically changing. A mobile application wouldn't require a dual core. Similarly a minimalist computer app like Zonbu would neither require a configuration that a gamer would demand. In these transitions there are specific lines of processors hitting the market. Likewise we might see Intel developing multi-core chips for servers and datacentres, and for mobile apps and minimal computing we might see power stingy microprocessors thronging the market.

There is Intel's Atom, the smallest and lowest power consuming processor from Intel. Tests suggest that an Atom based microprocessor exhausts about 32 watts of electricity; around 80 per cent more efficient, on the desktops and laptops of conventional users. Its widespread implementation would mean all the functions of a traditional desktop being catered at much lower costs. It really is worthwhile to let your CPU suck only as much as you require. There is no point in allowing your processor to consume 100 watts when all you would ever require for basic computing is 50 watts. Further advances would ensure that similar low power demanding tasks would be taken care by dedicated hardware designed specifically for that purpose resulting in significant efficiency. This clearly makes it evident that the processor as a 'central' unit will find diminished implementation in the time to come.

Microchips For Portables And Handhelds

In the move towards miniaturisation, it is predicted that PDAs and full-fledged mobile applications might well replace laptops. Even though that seems a bit further away, the total energy consumption by these devices, considering the sheer number of users it has got worldwide, cannot be overestimated. In this regards more energy efficient microchips that could lead to longer-lasting cell-phones and lesser recharge cycles is deemed necessary; especially when it is cognized that these devices amount to a sizeable proportion of electricity consumption.

One of the driving forces for any application has to be the million transistor packed silicon chips that power these devices.

